Nadya Farastanti

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Summary

Mechatronics Engineering student with hands-on expertise in electrical assembly, hardware integration, and industrial automation systems. Skilled in assembling precision mechanical components, electrical wiring, and integrating sensors, PLCs, and microcontrollers into functional engineering solutions. Detail-oriented and safety-focused, with strong technical troubleshooting skills for physical system prototyping, ready to contribute to the design of efficient and reliable systems in industrial environments.

Skills

- **Hardware & Assembly:** Mechanical & Electrical Assembly, Sensor Integration, Microcontrollers & PCB
- **Industrial Automation:** PLC (Schneider, Mitsubishi, Omron), MQTT
- **3D Design & CAD:** SolidWorks, CATIA

Experience & Selected Projects

Automated PPE Detection System with GPS Integration — Politeknik Negeri Batam
Final Year Thesis Project — *2026 – Present*

- Assembled hardware enclosure, GPS module, and embedded vision setup for real-time Personal Protective Equipment (PPE) detection using YOLOv26 algorithm.
- Designed and built a PyQt desktop interface featuring live video streaming, telemetry monitoring, and GPS data display.

Multi-Robot Communication & Base Station Integration System — Politeknik Negeri Batam
Internal Project — *2026*

- Developed and configured inter-robot wireless communication protocols and telemetry data transmission to a centralized base station interface.

AI-Based Wireless ECG — Politeknik Negeri Batam
Internal Project — *2025*

- Successfully designed and developed a wireless ECG monitoring prototype using the ADS1293 sensor and ESP32 microcontroller for real-time heart rate data acquisition.

PPE Detection System Using the YOLOv8 Algorithm — Politeknik Negeri Batam
Internal Project — *2025*

- Trained and deployed a YOLOv8 object detection model to classify 6 PPE types (helmets, safety glasses, masks, vests, gloves, and safety shoes).

PKM-KC (Student Innovation Program) — Kemendikbudristek
Student Innovation Project — *2024*

- Designed and developed a portable production metrics measuring tool integrated with Internet of Things (IoT) using the MQTT protocol.

Miniature Computer Integrated Manufacturing — Politeknik Negeri Batam
Internal Project — *2024*

- Integrated a robotic arm, conveyor belt, and SCARA robot to automate sorting and counting of metallic and non-metallic objects.

Education

Politeknik Negeri Batam — Batam, Indonesia
S.Tr. in Mechatronics Engineering — *Expected 2027*

Honors & Awards

Industrial Design Publication Certificate – ECG Project — DJKI
Intellectual Property Publication — *2026*

- Officially granted and published an Industrial Design certificate for the AI-Based Wireless ECG device enclosure and prototype.

Wirausaha Merdeka (WMK) Completion Certificate — Politeknik Negeri Batam
Entrepreneurship Flagship Program — *2024*

- Successfully developed "MICIGA" (Mini PLC Mega using Outseal V2 slim), designed as a compact, affordable, and user-friendly PLC module.

PKM-KC Grant Recipient Certificate Kemendikbudristek
Student Creativity Program – Innovation in Technology (PKM-KC). *2024*

- Awarded national funding grant for the portable IoT-based production metrics measurement system utilizing MQTT protocol.

Languages

- Indonesian (Native), English (Basic)